

Youngchan Kim

Ph.D. student @ MPI for Informatics

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🎓 Education

Max Planck Institute for Informatics

Ph.D. in Visual Computing and Artificial Intelligence

Apr 2026 – Present
Saarbrücken, Germany

POSTECH

M.S. in Artificial Intelligence

Mar 2022 – Feb 2024
Pohang, South Korea

Yonsei University

B.S. in Mathematics & Electrical and Electronic Engineering

Mar 2016 – Feb 2022
Seoul, South Korea

💻 Research Experience

Visual Computing and AI Department, MPI for Informatics

Doctoral Researcher

Apr 2026 – Present
Saarbrücken, Germany

- Advisors: Dr. Rishabh Dabral & Prof. Christian Theobalt
- Research Focus: (1) Embodied AI and World Models / (2) Human-Object-Scene Interaction

Computer Graphics Lab, POSTECH

Master's student

Mar 2022 – Feb 2024
Pohang, South Korea

- Advisor: Prof. Seung-Hwan Baek
- Research Focus: Multi-dimensional Light Properties Modeling

📚 Publications

* MULTI-DIMENSIONAL LIGHT PROPERTIES MODELING *

[1] Hyperspectral Polarimetric BRDFs of Real-world Materials

Yunseong Moon, Ryota Maeda, Suhyun Shin, Inseung Hwang, Youngchan Kim, Min H. Kim, Seung-Hwan Baek
SIGGRAPH Asia 2025

[2] Spectral and Polarization Vision: Spectro-polarimetric Real-world Dataset

Yujin Jeon*, Eunsue Choi*, Youngchan Kim, Yunseong Moon, Khalid Omer, Felix Heide, Seung-Hwan Baek
CVPR 2024 (highlight)

[3] Neural Spectro-polarimetric Fields

Youngchan Kim, Wonjoon Jin, Sunghyun Cho, Seung-Hwan Baek
SIGGRAPH Asia 2023

Work Experience

TDK Alight

ML Engineer - Synthetic Data Generation

Nov 2025 – Mar 2026

Remote

- Led the development of 3D face simulation pipeline for synthetic eye-gaze data generation

NationA

AI/ML Engineer

Aug 2025 – Nov 2025

Seoul, South Korea

- Implemented human motion synthesis and transfer models for web-based 3D content creation

DEEPX

AI Compiler Engineer

May 2025 – Jul 2025

Seongnam, South Korea

- Designed ONNX graph indexing and execution scheduling for efficient and flexible computation

Beeble AI

AI Researcher

Apr 2024 – May 2025

Seoul, South Korea

- Developed human-centric vision models for PBR map extraction and monocular depth estimation
- Built a virtual light stage in Blender for multi-view 3D human rendering with controllable lighting

Teaching Experience

Lecturer

- Data Analyst Bootcamp 12th - CODEIT Feb 2026 – Apr 2026
- Vibe Coding with Antigravity - ELICE Dec 2025
- AI Integration for Digital Transformation in Practice - KOSME Nov 2025
- Python and Data Analytics for Digital Transformation - SAMSUNG ELECTRONICS Apr 2024

Teaching Assistant

- AI Engineer Bootcamp 9th - CODEIT Dec 2025 – Apr 2026
- Deep Learning and Computer Vision - POSCO GROUP Oct 2022 / Jul 2023
- CS2700G TinCS: Computational Imaging - POSTECH Spring 2023

Academic Activities

REVIEWER Pacific Graphics 2023

Skills

LANGUAGES Korean (Native - 5/5), English (Conversational - 3~4/5)

PROGRAMMING & SW Python, PyTorch, TensorFlow, OpenCV, NumPy, Matplotlib, C/C++, Linux, Docker

Reference

Prof. Christian Theobalt

Ph.D. advisor

- Director, Visual Computing and AI Department, MPI for Informatics
- Professor, Department of Computer Science, Saarland University
- Director, Saarbrücken Center for Visual Computing, Interaction and Artificial Intelligence
- ✉ theobalt@mpi-inf.mpg.de | 🏠 <https://people.mpi-inf.mpg.de/~theobalt>

Dr. Rishabh Dabral

Ph.D. advisor

- Research Group Leader, 3D Visual Intelligence Research Group, MPI for Informatics
- ✉ rdabral@mpi-inf.mpg.de | 🏠 <https://rishabhdabral.github.io>

Prof. Seung-Hwan Baek

M.S. advisor

- Associate Professor, Department of Computer Science and Engineering, POSTECH
- Affiliated Professor, Graduate School of Artificial Intelligence, POSTECH
- ✉ shwbaek@postech.ac.kr | 🏠 <https://www.shbaek.com>